

The Phase Transition Theorem: When Emergence Is Real, and When It Is a Metric Artifact

(Synthesis of the stationary-barrier, abstraction, and spectral-benchmark programmes)

Ivan Drozdov

Abstract

The three companion manuscripts each describe one face of the same event. TR characterises the world *before* a representational transition: a stationary feature-learning attractor that self-organises to source exponent $r = \frac{1}{2}$ and is permanently pinned to the Sobolev minimax barrier $\beta_0 = 2s/(2s + d^*)$. AB supplies the exact objects that characterise the *instant* of transition: the defect $\delta(\phi, \mathcal{F})$, which sees a feature map only through the subspace V_ϕ it generates; the subspace gap $\Theta(V_1, V_2)$; the compression frontier whose exponent *is* the effective dimension d_{loc}^* ; and the interaction gap Δ as the order parameter that gates the collapse of the defect. BM shows the transition occurring on a controlled synthetic manifold: the compositional approximation exponent jumps to $2s/d_{\text{loc}}^*$ while the Sobolev data-scaling barrier stays at β_0 . We unify the three into a single statement. We prove the existence of a critical parameter scale N_c , finite and independent of the sample budget D , that separates a *subcritical* stationary regime (loss exponents $\alpha = 2s/d^*$, $\beta = \beta_0$) from a *supercritical* compositional regime (exponents $\alpha' = 2s/d_{\text{loc}}^*$, $\beta' = 2s/(2s + d_{\text{loc}}^*) > \beta_0$), with the binding effective dimension dropping from d^* to d_{loc}^* as V_ϕ snaps onto the width-optimal subspace V_{opt} . We show the transition is *sharp* in the $D \rightarrow \infty$ limit, modulo a single named bifurcation hypothesis inherited verbatim from TR. Finally we prove an *authenticity criterion*: the jump is strictly positive in the L^2 metric *if and only if* $d_{\text{loc}}^* < d^*$; when $d_{\text{loc}}^* = d^*$ no L^2 transition exists, and any observed jump under a nonlinear metric is a metric artifact. This is the exact discriminating invariant the Schaeffer–Wei debate lacked: emergence is real precisely when the target carries genuine compositional structure ($d_{\text{loc}}^* < d^*$), a property measurable from the compression-frontier exponent.

Index Terms

emergent abilities, scaling laws, feature learning, phase transitions, effective dimension, nonlinear width, minimax rates, bifurcation theory.

I. INTRODUCTION

Two readings of *emergence* in large models stand opposed. One holds that sharp capability jumps are mirages produced by discontinuous or nonlinear evaluation metrics: under a smooth metric the same curves are smooth [4]. The other holds that the jumps are real features of the learned representation, not of the ruler [5]. The debate has lacked a mathematical object that decides which reading applies to a given task. This paper supplies one.

The decision is made by a single scalar invariant of the concept class: its *effective dimension* d_{loc}^* , the exponent of the compression frontier of AB (theorem II.1 (c)). We prove that a genuine, metric-independent phase transition in the loss exponents exists exactly when $d_{\text{loc}}^* < d^*$, and that when $d_{\text{loc}}^* = d^*$ the only transitions available are metric artifacts.

The connection between the transition’s existence and d_{loc}^* was not obvious from the outset. The benchmark BM showed the jump numerically on a flat torus before the analytic picture was clear. Working

backward from that experiment—asking why the jump appeared in one synthetic target and not another—eventually isolated $\Delta = \sigma_{\min}(\mathbb{E}[\nabla^2 g]) > 0$ as the gating quantity. The present paper is the analytic explanation of what the benchmark was showing.

The three companion manuscripts each furnish one ingredient of the proof.

a) The division of labour. TR is the theory of the subcritical phase. Its stationary attractor, its self-organised criticality $r = \frac{1}{2}$, and its barrier β_0 are the exact mathematical characterisation of the regime in which the representation has *not yet* captured compositional structure. AB is the theory of the transition itself: the defect $\delta(\phi, \mathcal{F}) = \sup_f \|(I - P_\phi)f\|$ is a projection residual depending on ϕ only through V_ϕ , so a sharp fall of the defect coincides with the event “ V_ϕ aligns with the optimal subspace V_{opt} ,” and the gap Δ is the order parameter that gates it. BM is the proof of concept: on the flat torus with a planted compositional target, the approximation exponent jumps to $2s/d_{\text{loc}}^* = 2.5$ while the data barrier stays at β_0 . Our task is to lift this from a controlled instance to a theorem.

b) What we prove, and at what grade. theorem IV.1 (P.i), (P.ii) and (P.iv) are theorems under stated hypotheses, assembled from results already proved in the three sources. theorem IV.1 (P.iii) (sharpness) rests on one named bifurcation hypothesis (H-bif) (theorem VI.1): it is the transfer of the saddle-to-saddle structure of [6] to the manifold/Langevin setting, the single residual hypothesis already present in TR. We flag exactly where it is used.

II. SETUP

We adopt the common data model of the three sources.

Definition II.1 (The unified setting).

- (a) Manifold and basis. Inputs $x \in \mathbb{R}^d$ lie on a compact Riemannian manifold $\mathcal{M}$ of intrinsic dimension $d^* \ll d$, with measure μ and Laplace–Beltrami eigenbasis $-\Delta_{\mathcal{M}}\psi_k = \mu_k\psi_k$, $\mu_k \asymp k^{2/d^*}$ (Weyl). The order- s Sobolev kernel has eigenvalues $\sigma_k \asymp k^{-b_0}$, $b_0 = 2s/d^*$.
- (b) Concept class. $\mathcal{F} = \mathcal{C}_{d_{\text{loc}}^*}^s$ is the compositional Sobolev class: targets $f^* = g_L \circ \dots \circ g_1$ with each constituent $g_i \in \mathcal{H}^s$ depending on an intermediate representation of intrinsic dimension d_i , and $d_{\text{loc}}^* = \max_i d_i \leq d^*$. (For $L = 1$, or any full-dimensional constituent, $d_{\text{loc}}^* = d^*$ and $\mathcal{F}$ is an ordinary Sobolev ball.)
- (c) Effective dimension as a frontier exponent (AB). The constrained nonlinear m -width (compression frontier) of $\mathcal{F}$ under a bounded-capacity decoder class $\mathcal{G}$ satisfies $\delta_m^*(\mathcal{F}; \mathcal{G})^2 \asymp m^{-2s/d_{\text{loc}}^*}$, and this defines d_{loc}^* . For a Sobolev ball $d_{\text{loc}}^* = d^*$; only genuine composition gives $d_{\text{loc}}^* < d^*$.
- (d) Feature maps. Parametric feature maps $\phi_\theta : \mathcal{X} \rightarrow \mathcal{Z}$ with $|\theta| = N$ parameters, realised by an L -layer μP network with smooth bounded ϕ ($\phi'' \in L^\infty$). Write V_ϕ for the closed subspace $L^2(\mathcal{X}, \sigma(\phi), \mu)$ and P_ϕ for its orthogonal projection.
- (e) Data. D i.i.d. samples $(x_i, f^*(x_i) + \xi_i)$ of a single $f^* \in \mathcal{F}$, sub-Gaussian noise ξ_i .

We will need three named objects from the sources, recalled here so the paper is self-contained.

Definition II.2 (Optimal subspace, defect, subspace gap).

- The width-optimal subspace $V_{\text{opt}} = V_m(\mathcal{F})$ is the m -dimensional closed subspace attaining the linear/manifold width of $\mathcal{F}$ at resolution m ; for $\mathcal{C}_{d_{\text{loc}}^*}^s$ with $d_{\text{loc}}^* < d^*$ it is the span of the d_{loc}^* -dimensional compositional flag, not the ambient eigenbasis (AB, parallel-existence flag).
- The defect $\delta(\phi, \mathcal{F}) = \sup_{f \in \mathcal{F}} \|(I - P_\phi)f\|_{L^2(\mu)}$, which by the projection identity of AB depends on ϕ only through V_ϕ .
- The subspace gap $\Theta(V_1, V_2)$ is the operator (Kato) gap between equidimensional closed subspaces; $\Theta = 0$ iff $V_1 = V_2$, $\Theta = 1$ iff they are orthogonal.
- The interaction gap $\Delta = \sigma_{\min}(\mathbb{E}[\nabla^2 g])$ (level-2 instance) and $\Delta_{\min} = \min_i \Delta_i$ along the chain; by AB the gap is simultaneously the existence, factorisation, and learnability order parameter, with the statistical-query phase boundary at $\Delta_{\min} = 0$.

The single quantity that the whole transition is *about* is the binding effective dimension.

Definition II.3 (Binding dimension). *For a feature map ϕ with subspace V_ϕ , the binding dimension $d_b(\phi)$ is the exponent governing the slowest-decaying residual at the operating point: $d_b = d^*$ when V_ϕ is misaligned with the compositional flag ($\Theta(V_\phi, V_{\text{opt}})$ bounded away from 0), and $d_b = d_{\text{loc}}^*$ when $V_\phi \approx V_{\text{opt}}$ ($\Theta(V_\phi, V_{\text{opt}}) \rightarrow 0$). The realised loss exponents are then $\alpha = 2s/d_b$ and $\beta = 2s/(2s + d_b)$, the per-dimension copies of the TR identities.*

III. THE TWO ATTRACTORS

The mean-field closure of TR (its Proposition on Yang–Hu closure) reduces the evolving NTK to a spectral gradient system

$$\frac{d\lambda_k}{dt} = -\eta \frac{\partial V}{\partial \lambda_k} + \Gamma_k[\{\lambda_j\}_{j \neq k}, t], \quad V(\boldsymbol{\lambda}) = \sum_k \frac{\bar{a}_k^2}{\lambda_k} + \mu \sum_k \lambda_k^\nu, \quad (1)$$

with capacity exponent $\nu = 1/L$ and target coefficients $\bar{a}_k$. We record the two stationary points relevant to the transition; both are special cases of the TR attractor, differing only in which subspace carries the target energy.

Lemma III.1 (Subcritical (stationary) attractor). *Suppose V_ϕ is misaligned with the compositional flag, $\Theta(V_\phi, V_{\text{opt}}) \geq \theta_0 > 0$, so the target energy is spread across a full- d^* block. Then the KKT condition on (1) gives $\lambda_k^* \asymp k^{-b}$ with $b = (1 + 2t)/(\nu + 1)$, the realised source exponent is $r(\nu) = t(\nu + 1)/(1 + 2t)$ with ceiling $r = \frac{1}{2}$ at $\nu_\star = 1/(2t)$, and the loss obeys*

$$L(N, D) \asymp N^{-\alpha} + D^{-\beta_0} + L_0, \quad \alpha = \frac{2s}{d^*}, \quad \beta_0 = \frac{2s}{2s + d^*}. \quad (2)$$

Proof. This is TR read at binding dimension d^* : the attractor and the ceiling $r = \frac{1}{2}$ are its self-organised-criticality theorem; the estimation exponent at $r = \frac{1}{2}$ is $\beta(\frac{1}{2}) = \beta_0$ via the source/capacity rate $\beta(r) = 4rs/(4rs + d^*)$; the approximation exponent $\alpha = 2s/d^*$ is its no-adaptivity-gain theorem on the Sobolev ball, valid because a misaligned V_ϕ sees only the ambient width ($d_{\text{loc}}^*(V_\phi) = d^*$ by theorem II.1 (c) restricted to V_ϕ). $\square$

Lemma III.2 (Supercritical (compositional) attractor). *Suppose $V_\phi \approx V_{\text{opt}}$, $\Theta(V_\phi, V_{\text{opt}}) \leq \theta_1 \ll 1$, and the chain gap is open, $\Delta_{\min} > 0$. Then the defect collapses to the d_{loc}^* -floor, $\delta(\phi, \mathcal{F})^2 \asymp m^{-2s/d_{\text{loc}}^*}$, the same stationarity analysis run at binding dimension d_{loc}^* gives $r = \frac{1}{2}$ relative to the d_{loc}^* -effective kernel, and*

$$L(N, D) \asymp N^{-\alpha'} + D^{-\beta'} + L_0, \quad \alpha' = \frac{2s}{d_{\text{loc}}^*}, \quad \beta' = \frac{2s}{2s + d_{\text{loc}}^*} > \beta_0. \quad (3)$$

Proof. By the projection identity of AB, $V_\phi \approx V_{\text{opt}}$ makes the defect the width frontier of $\mathcal{F}$, which has exponent $2s/d_{\text{loc}}^*$ by theorem II.1 (c); this is the approximation term N^{-2s/d_{loc}^*} once N parameters resolve the flag (AB parallel-existence theorem, dominated by the hardest level $d_{\text{loc}}^* = \max_i d_i$). The estimation term is the TR analysis with $d^* \mapsto d_{\text{loc}}^*$: the matched-depth attractor at the effective dimension again realises $r = \frac{1}{2}$, now against the d_{loc}^* -kernel, so $\beta' = 2s/(2s + d_{\text{loc}}^*)$. Positivity of $\Delta_{\min}$ is what makes V_{opt} recoverable and the link identifiable (AB clause-(b)=gap). The strict inequality $\beta' > \beta_0$ is the elementary $2s/(2s + d_{\text{loc}}^*) - 2s/(2s + d^*) = 2s(d^* - d_{\text{loc}}^*)/[(2s + d_{\text{loc}}^*)(2s + d^*)] > 0$ for $d_{\text{loc}}^* < d^*$. $\square$

Remark III.3 (Both attractors sit at $r = \frac{1}{2}$). *The transition is not a jump in the source exponent r . Both phases are stationary ($r = \frac{1}{2}$); what jumps is the binding dimension entering $\beta_0(d) = 2s/(2s + d)$, as V_ϕ snaps from a full- d^* block onto the d_{loc}^* -flag. This is the “resource/structure crossing leaves r flat but moves the binding dimension” phenomenon of TR’s two-staircases remark, now promoted to the engine of an emergence event.*

IV. THE PHASE TRANSITION THEOREM

Theorem IV.1 (Phase transition). *Under theorem II.1, with a chain gap $\Delta_{\min} > 0$ when $d_{\text{loc}}^* < d^*$, there exists a critical parameter scale $N_c = N_c(s, d^*, d_{\text{loc}}^*, \Delta_{\min})$, finite and independent of D , such that:*

(P.i) Subcritical $N < N_c$. *The optimal ϕ_N is the stationary attractor of theorem III.1: $\Theta(V_{\phi_N}, V_{\text{opt}}) \geq \theta_0 > 0$, $r(\nu^*) = \frac{1}{2}$, and*

$$L(N, D) \asymp N^{-\alpha} + D^{-\beta_0} + L_0, \quad \alpha = \frac{2s}{d^*}, \quad \beta_0 = \frac{2s}{2s+d^*}.$$

(P.ii) Supercritical $N > N_c$. *The optimal ϕ_N is the compositional attractor of theorem III.2: $V_{\phi_N} \approx V_{\text{opt}}$ (so $\Theta \rightarrow 0$ and the defect attains the d_{loc}^* -nonlinear width), and*

$$L(N, D) \asymp N^{-\alpha'} + D^{-\beta'} + L_0, \quad \alpha' = \frac{2s}{d_{\text{loc}}^*}, \quad \beta' = \frac{2s}{2s+d_{\text{loc}}^*} > \beta_0.$$

(P.iii) Sharpness. *Under (H-bif) (theorem VI.1), in the limit $D \rightarrow \infty$ the data exponent $\beta(N)$ is discontinuous at N_c :*

$$\lim_{D \rightarrow \infty} [\beta(N_c + \varepsilon) - \beta(N_c - \varepsilon)] \geq \Delta_\beta := \frac{2s(d^* - d_{\text{loc}}^*)}{(2s + d_{\text{loc}}^*)(2s + d^*)} > 0, \quad \forall 0 < \varepsilon \ll 1. \quad (4)$$

(P.iv) Authenticity criterion. *The jump is strictly positive in the L^2 metric iff $d_{\text{loc}}^* < d^*$. If $d_{\text{loc}}^* = d^*$ then $\Delta_\beta = 0$, no L^2 transition exists, the defect $\delta(\phi_N, \mathcal{F})$ is smooth in N along the single frontier N^{-2s/d^*} , and any discontinuity observed under a nonlinear metric $M \neq L^2$ is a metric artifact in the sense of theorem VII.1.*

We prove the four parts in the three technical steps named in the problem statement: existence of N_c (section V), sharpness (section VI), and the authenticity criterion (section VII). (P.i)/(P.ii) are theorems III.1 and III.2 once N_c is in hand.

V. STEP 1: EXISTENCE OF THE CRITICAL SCALE

The key structural fact is that the compositional attractor is not merely energetically preferred above some crossover; it is *unavailable* below a threshold, because the network cannot realise the d_{loc}^* -flag with an open gap until it has enough parameters. The threshold is a capacity threshold, hence D -independent, which is what makes the $D \rightarrow \infty$ limit of (P.iii) well-posed.

Proposition V.1 (Realisability threshold). *Let $d_{\text{loc}}^* < d^*$ and $\Delta_{\min} > 0$. There is a finite $N_c = N_c(s, d^*, d_{\text{loc}}^*, \Delta_{\min})$, independent of D , such that:*

- (i) *for $N < N_c$ no realisable ϕ_N satisfies both $\Theta(V_{\phi_N}, V_{\text{opt}}) \leq \theta_1$ and an open realised gap $\hat{\Delta}_i > 0$ for all i ; the compositional attractor of theorem III.2 is not a critical point of (1) in the realisable set, so the optimiser rests on the stationary attractor of theorem III.1;*
- (ii) *for $N > N_c$ such a ϕ_N exists and the compositional attractor is the strictly lower-loss critical point ($d_{\text{loc}}^* < d^*$);*
- (iii) *the leading scaling is $N_c \asymp \Delta_{\min}^{-d_{\text{loc}}^*/s}$ up to constants depending on (s, d_{loc}^*) and the decoder capacity.*

Proof. (i)–(ii). Realising the binding level of V_{opt} to defect within the d_{loc}^* -floor requires, by the compositional/parallel-existence rate of AB, resolving the hardest constituent (local dimension d_{loc}^*) to some fixed accuracy ε_b , at parameter cost $N \gtrsim \varepsilon_b^{-d_{\text{loc}}^*/s}$ (the Sobolev rate in its own dimension). The accuracy at which the realised level- i gap $\hat{\Delta}_i = \Delta_i[\hat{\nu}_{i-1}]$ stays open is, by the gap-stability lemma of AB, $\hat{\Delta}_i \geq \Delta_i - L_i \varepsilon$; an open gap therefore demands $\varepsilon_b \lesssim \Delta_{\min} / \max_i L_i$. Substituting fixes a finite parameter floor N_c . Below it, either V_ϕ cannot reach $\Theta \leq \theta_1$ or some realised gap closes; in either case theorem III.2's hypotheses fail and its attractor is not a critical point of (1) on the realisable manifold, leaving theorem III.1's stationary attractor as the operating point. Above it both hypotheses hold and, since $\beta' > \beta_0$ and $\alpha' > \alpha$ for $d_{\text{loc}}^* < d^*$, the compositional critical point has strictly lower loss at every (N, D) . (iii). Eliminating

ε_b between $N \asymp \varepsilon_b^{-d_{\text{loc}}^*/s}$ and $\varepsilon_b \asymp \Delta_{\min}/\max_i L_i$ gives $N_c \asymp (\max_i L_i/\Delta_{\min})^{d_{\text{loc}}^*/s} \asymp \Delta_{\min}^{-d_{\text{loc}}^*/s}$. None of these estimates involves D . $\square$

Remark V.2 (Why N_c does not depend on D). N_c is set by expressivity (can the network instantiate the flag with an open gap?), not by statistics (how many samples to estimate the coordinates). The sample budget enters only the estimation term $D^{-\beta}$; it moves the loss along whichever branch is active but does not create or destroy the compositional branch. This separation is what allows us to send $D \rightarrow \infty$ at fixed N_c in (P.iii).

Theorem V.1 gives (P.i) and (P.ii) immediately: below N_c the operating point is theorem III.1 (subcritical exponents); above N_c it is theorem III.2 (supercritical exponents).

VI. STEP 2: SHARPNESS OF THE TRANSITION

This is the hardest step: showing that the stationary attractor does not merely become suboptimal at N_c but *loses stability*, so that V_{ϕ_N} snaps onto V_{opt} over a window that shrinks to a point as $D \rightarrow \infty$. We isolate the one hypothesis this requires and then prove the jump.

Assumption VI.1 ((H-bif): the leap is a hyperbolic bifurcation). *At $N = N_c$ the stationary attractor of theorem III.1 is a hyperbolic equilibrium of the mean-field flow (1) whose loss of stability, as N crosses N_c , occurs in exactly the d_{loc}^* -dimensional mode subspace of the binding compositional constituent; equivalently, the unstable manifold born at N_c has dimension d_{loc}^* and coincides with the leap of the saddle-to-saddle picture.*

Remark VI.2 ((H-bif) is inherited, not new). *TR already derives, by a per-level Freidlin–Wentzell/Kramers analysis, that a metastable plateau is a hyperbolic saddle whose unstable manifold is the d_i -dimensional subspace of the constituent being resolved, with the leap complexity equal to the local dimension. The only thing TR leaves open is the transfer of the hypercube saddle-to-saddle structure of [6] to the manifold/Langevin setting — a named residual hypothesis that bears on dynamics and constants, not on the asymptotic rate. (H-bif) is precisely that residual, no more. No assumption beyond what TR has already itemised is added.*

Proposition VI.3 (Stability exchange at N_c). *Linearise (1) about the stationary attractor and restrict to the d_{loc}^* -dimensional compositional subspace. The relevant eigenvalue of the Hessian of $V_{\text{tot}} = V + \Phi$ has the form*

$$\omega(N) \asymp c_1 \Delta_{\min}^2 - c_2 \rho(N), \quad \rho(N) \downarrow 0 \text{ as } N \uparrow \infty, \quad (5)$$

where $\rho(N)$ is the realisability deficit (the residual misalignment energy of the best realisable V_ϕ at budget N) and $c_1, c_2 > 0$ depend only on (s, d_{loc}^*) . Under (H-bif), ω crosses zero transversally at a single N_c : the stationary attractor is stable ($\omega < 0$) for $N < N_c$ and a saddle ($\omega > 0$) for $N > N_c$, with N_c the root of (5) matching theorem V.1.

Proof. The reconstruction curvature pins the diagonal block of the Hessian; the compositional off-diagonal coupling enters through the level-2 score tensor, whose conditioning is exactly the gap Δ (AB clause-(b)=gap), so the restoring term in the compositional directions scales as $\Delta_{\min}^2$. The destabilising term is the realisability deficit $\rho(N)$: when N is too small to instantiate the flag, the misaligned configuration is locally stabilised by the capacity penalty, contributing $-c_2 \rho(N)$. As N grows, $\rho(N) \rightarrow 0$ (the flag becomes representable), and the sign of ω is eventually set by the positive $c_1 \Delta_{\min}^2$. (H-bif) provides hyperbolicity and a d_{loc}^* -dimensional unstable manifold, giving a transversal single crossing at N_c rather than a degenerate or higher-codimension one. The crossing N coincides with the realisability threshold of theorem V.1 because both are governed by $\rho(N_c) \asymp \Delta_{\min}^2$. $\square$

We now isolate exactly what part of (H-bif) is automatic and what part is the genuine residual. The transversality of the crossing is unconditional; the codimension and the dimension of the unstable manifold reduce to two named spectral conditions on the mean-field Hessian.

Lemma VI.4 (Conditional reduction of (H-bif)). *Write the Hessian of V_{tot} at the stationary attractor, in the eigenbasis that diagonalises it away from the transition, as $\text{Hess} = \text{diag}(\kappa_1(N), \kappa_2(N), \dots) \oplus B(N)$, where $B(N)$ is the block spanned by the d_{loc}^* compositional (binding) modes and $\{\kappa_k(N)\}$ are the bulk curvatures. Assume the realisability deficit is the parent rate $\rho(N) \asymp N^{-2s/d_{\text{loc}}^*}$, strictly decreasing and C^1 in N . Suppose:*

- (H-gap) *there is $\kappa_0 > 0$ with $\kappa_k(N) \geq \kappa_0$ for all bulk modes k and all N in a neighbourhood of N_c (uniform bulk spectral gap); and*
- (H-block) *the destabilising contribution $-c_2\rho(N)$ enters only the binding block $B(N)$, whose smallest eigenvalue is $\omega(N) \asymp c_1\Delta_{\text{min}}^2 - c_2\rho(N)$ of (5).*

Then: (i) ω has a single zero N_c and the crossing is transversal, $\omega'(N_c) > 0$; (ii) at N_c exactly one eigenvalue changes sign, so the loss of stability is codimension one; and (iii) the unstable manifold born at N_c has dimension d_{loc}^ and is the binding block. Consequently (H-gap) + (H-block) $\Rightarrow$ (H-bif).*

Proof. (i) By (H-block), $\omega(N) = c_1\Delta_{\text{min}}^2 - c_2\rho(N)$ up to the asymptotic equivalence, and ρ is strictly decreasing and C^1 , so ω is strictly increasing and C^1 ; hence it has at most one zero, and $\omega'(N_c) = -c_2\rho'(N_c) > 0$ since $\rho' < 0$. The zero exists because $\omega(N) \rightarrow c_1\Delta_{\text{min}}^2 > 0$ as $N \rightarrow \infty$ and $\omega(N) < 0$ for small N (where the deficit dominates), giving the transversal crossing. (ii) By (H-gap) every bulk curvature stays $\geq \kappa_0 > 0$ across the neighbourhood, so no bulk eigenvalue can vanish at N_c ; the only eigenvalue reaching zero is the one in $B(N)$ from (i), so the sign change is codimension one. (iii) For $N > N_c$ the only negative eigendirections are those of $B(N)$, which has rank d_{loc}^* , so the unstable manifold has dimension d_{loc}^* and equals the binding block. This is the content of (H-bif). $\square$

Remark VI.5 (What the reduction does and does not settle). *Theorem VI.4 replaces the single opaque hypothesis (H-bif) by two spectral conditions on the mean-field Hessian: a uniform bulk gap (H-gap) and the confinement of the destabilisation to the binding block (H-block). The transversality (i) is now a theorem — it follows solely from the monotone C^1 realisability deficit $\rho(N) \asymp N^{-2s/d_{\text{loc}}^*}$ and needs no dynamical input. What is not discharged is (H-gap) and (H-block) themselves: proving them for the actual Langevin/DMFT flow amounts to the transfer of the saddle-to-saddle structure of [6] to the manifold setting (theorem VI.2), and a hand-built reduced potential that postulates a bounded-below bulk is an illustration, not a proof, of (H-gap). The value of the lemma is to pinpoint that the entire residual in (P.iii) is the pair {uniform bulk gap, single binding block}; the sharpness clause remains conditional on them, while the exponents and the criterion (P.iv) do not depend on them at all.*

Proof of theorem IV.1 (P.iii). By theorem VI.3, for $N < N_c$ the stationary attractor is asymptotically stable, so the optimiser remains there for all D and, by theorem III.1, $\beta(N) = \beta_0$. For $N > N_c$ it is a saddle whose d_{loc}^* -dimensional unstable manifold (the leap, by (H-bif)) carries the flow to the compositional attractor of theorem III.2, so $\beta(N) = \beta'$. For finite D the realised exponent near N_c is blurred over a window of width $w(D)$ set by the estimation noise that competes with the eigenvalue $\omega(N)$; standard escape-time/relaxation analysis gives $w(D) \rightarrow 0$ as $D \rightarrow \infty$ (the noise floor that hides the unstable direction vanishes when the coordinates are resolved). Hence

$$\lim_{D \rightarrow \infty} \beta(N_c + \varepsilon) = \beta' = \frac{2s}{2s + d_{\text{loc}}^*}, \quad \lim_{D \rightarrow \infty} \beta(N_c - \varepsilon) = \beta_0 = \frac{2s}{2s + d^*},$$

and their difference is $\Delta_\beta = 2s(d^* - d_{\text{loc}}^*)/[(2s + d_{\text{loc}}^*)(2s + d^*)] > 0$ for every $0 < \varepsilon \ll 1$, which is (4). $\square$

Remark VI.6 (The two staircases, distinguished). *theorem VI.3 sharpens TR's distinction between a resource crossing (static, r stays $\frac{1}{2}$, no instability) and a dynamical saddle escape (r rises transiently). The emergence event here is the second kind as a function of N but lands on a new stationary attractor that is again $r = \frac{1}{2}$: the system leaps from one self-organised-critical point to another, and only the binding dimension records the leap. This is why TR's decisive test (r flat across a kink) correctly reads the post-transition regime as stationary even though a genuine instability was crossed in N .*

VII. STEP 3: THE AUTHENTICITY CRITERION

We now prove (P.iv), the half of the theorem that closes the Schaeffer–Wei debate. The content is that the existence of a metric-independent jump is equivalent to $d_{\text{loc}}^* < d^*$, and that the $d_{\text{loc}}^* = d^*$ case is the regime where apparent jumps are artifacts of a nonlinear ruler.

Definition VII.1 (Metric artifact). *A performance metric is a map $M = \psi \circ \delta_{L^2}$ for some nondecreasing $\psi : [0, \infty) \rightarrow [0, \infty)$. A jump in $M(N)$ at N_0 is a metric artifact if $\delta_{L^2}(\cdot, \mathcal{F})$ is continuous (indeed smooth) at N_0 ; equivalently, the jump does not survive passage to a bi-Lipschitz-equivalent metric. A jump is genuine if it appears in δ_{L^2} itself, hence under every faithful (bi-Lipschitz) metric.*

Lemma VII.2 ($d_{\text{loc}}^* = d^*$ kills the L^2 transition). *If $d_{\text{loc}}^* = d^*$ then V_{opt} is (rate-equivalent to) the ambient Laplacian eigenbasis, the subspace gap $\Theta(V_{\phi_N}, V_{\text{opt}})$ does not jump, and $\delta(\phi_N, \mathcal{F})^2 \asymp N^{-2s/d^*}$ is the single smooth frontier for all N . In particular $\beta(N) \equiv \beta_0$ and $\Delta_\beta = 0$.*

Proof. $d_{\text{loc}}^* = d^*$ is, by AB’s width theorem and TR’s no-adaptivity-gain theorem, exactly the statement that the linear and nonlinear N -widths of $\mathcal{F}$ coincide in rate: sorting the coefficients buys no exponent, the worst-case sorted decay equals the linear decay $\tilde{t} = t = s/d^*$, and the eigenbasis is already rate-optimal. Hence there is no distinguished low-dimensional flag for V_ϕ to align onto: any rate-optimal V_{ϕ_N} is width-equivalent to the ambient block, so $\Theta(V_{\phi_N}, V_{\text{opt}})$ carries no information and cannot collapse. The defect is the single Sobolev frontier N^{-2s/d^*} , real-analytic in N . With no second attractor available (theorem III.2 requires $d_{\text{loc}}^* < d^*$ to give a strictly lower loss), theorem VI.3’s eigenvalue never changes sign — $c_1 \Delta_{\min}^2$ no longer gates a *dimension reduction* because there is none — so the stationary attractor is globally stable and $\beta(N) \equiv \beta_0$. $\square$

Proof of theorem IV.1 (P.iv). ($\Leftarrow$) If $d_{\text{loc}}^* < d^*$, (P.iii) exhibits a strictly positive jump $\Delta_\beta > 0$ in β , which is a property of the L^2 risk exponents and hence genuine. ($\Rightarrow$) If $d_{\text{loc}}^* = d^*$, theorem VII.2 gives a smooth L^2 defect and $\Delta_\beta = 0$: no L^2 transition. Suppose a metric $M = \psi \circ \delta_{L^2}$ nonetheless reports a sharp jump at some N_0 — for instance a thresholded/exact-match metric $\psi(u) = \mathbb{1}[u < \tau]$, the canonical discontinuous evaluation of [4]. Since δ_{L^2} is smooth at N_0 , the jump in M comes entirely from the discontinuity of ψ , not from the representation: it is a metric artifact by theorem VII.1, and is removed by any bi-Lipschitz reparametrisation of the metric. Thus a genuine (L^2 , metric-independent) jump exists *iff* $d_{\text{loc}}^* < d^*$. $\square$

VIII. CLOSING THE DEBATE

Theorem IV.1 adjudicates the two camps by exhibiting the missing invariant.

Corollary VIII.1 (The discriminating invariant). *Emergence in the sense of a metric-independent capability jump is real for a concept class $\mathcal{F}$ if and only if $d_{\text{loc}}^*(\mathcal{F}) < d^*$, where d_{loc}^* is the exponent of $\mathcal{F}$ ’s compression frontier. When $d_{\text{loc}}^* = d^*$, every apparent jump is a metric artifact. The invariant $d_{\text{loc}}^* < d^*$ is checkable from data by the two decisive tests below.*

	[4]	[5]	This theorem
Is emergence real?	No	Yes	Iff $d_{\text{loc}}^* < d^*$
Mathematical criterion?	—	—	Subspace gap / frontier exponent
Explains artifacts?	Partially	No	Yes (P.iv, $d_{\text{loc}}^* = d^*$)

Both camps are right, conditionally: Schaeffer’s mirage is the $d_{\text{loc}}^* = d^*$ regime (smooth L^2 defect, discontinuous ruler), and Wei’s real jump occurs in the $d_{\text{loc}}^* < d^*$ regime (genuine subspace realignment with an open gap).

a) Synthetic instantiation (BM).: On the flat torus $\mathbb{T}^{d^*}$ with $d^* = 2$, $s = 1.25$, and a compositional target with $d_{\text{loc}}^* = 1$, the theorem predicts $\beta_0 = 2.5/4.5 \approx 0.556$, $\beta' = 2.5/3.5 \approx 0.714$, hence a genuine jump $\Delta_\beta \approx 0.159$, alongside the approximation jump $\alpha = 2s/d^* = 1.25 \rightarrow \alpha' = 2s/d_{\text{loc}}^* = 2.5$. BM reports exactly the compositional approximation exponent $2s/d_{\text{loc}}^* = 2.5$ against a Sobolev exponent held at β_0 — the controlled-manifold signature of a $d_{\text{loc}}^* < d^*$ transition, i.e. a genuine event in the sense of (P.iv).

b) Decisive tests (TR/BM protocol).: To certify a transition in a trained model: (i) estimate d^* from the data manifold (graph-Laplacian spectrum) and d_{loc}^* from the model-side approximation exponent $\alpha = 2s/d_{\text{loc}}^*$; a strict double-dimension consistency $d_{\text{loc}}^* < d^*$ certifies a genuine transition. (ii) Read the source exponent $\hat{r}$ across widths and across the kink: by theorem VI.6 $\hat{r}$ flat at $\frac{1}{2}$ on both sides marks the leap between two self-organised-critical attractors (genuine, $d_{\text{loc}}^* < d^*$), whereas a smooth L^2 curve under a discontinuous metric marks an artifact ($d_{\text{loc}}^* = d^*$).

IX. STATUS AND GRADING

- **Theorems (under stated hypotheses).** theorems III.1 and III.2 (the two attractors, from TR’s criticality and AB’s width frontier); theorem V.1 and hence (P.i), (P.ii) (existence of a finite, D -independent N_c , from AB’s gap-stability and parallel-existence results); theorem VII.2 and hence (P.iv) (the authenticity criterion, from the linear=nonlinear width coincidence at $d_{\text{loc}}^* = d^*$).
- **Conditional on one named hypothesis, now reduced to two spectral conditions.** theorem VI.3 and (P.iii) (sharpness) rest on (H-bif), the hyperbolic-bifurcation/leap structure, which is the saddle-to-saddle transfer hypothesis already isolated in TR (theorem VI.2). Theorem VI.4 sharpens this: the *transversality* of the crossing is now unconditional (it follows from the monotone C^1 realisability deficit), and the residual is exactly the pair {uniform bulk spectral gap (H-gap), single binding block (H-block)}. The *asymptotic exponents* $\beta_0, \beta', \alpha, \alpha'$ and the criterion (P.iv) do not depend on it; only the *sharpness* of the crossover does.
- **The single load-bearing invariant.** Everything reduces to one inequality, $d_{\text{loc}}^* < d^*$, equivalently the non-vanishing of a subspace gap / interaction gap. This is the object AB isolates as the common reading of existence, factorisation, learnability, and the statistical-query transition, and it is here the reading of *emergence* as well.

a) Conclusion.: The three manuscripts compose into one event seen at three times. TR gives the stable subcritical attractor and its barrier; AB gives the exact objects of the transition — the defect as a projection, the subspace gap, the effective dimension, and the gap as order parameter; BM shows the event on a controlled manifold. The Phase Transition Theorem ties them: a genuine, metric-independent emergence occurs at a finite, D -independent parameter scale N_c precisely when the target’s effective dimension is below ambient, $d_{\text{loc}}^* < d^*$, and is otherwise a metric artifact. The open work is to discharge the two spectral conditions (H-gap) and (H-block) to which theorem VI.4 has reduced (H-bif) — concretely, to prove a uniform spectral gap for the bulk of the mean-field Hessian and the confinement of the destabilising deficit to the d_{loc}^* -dimensional binding block, by transporting the saddle-to-saddle structure of [6] to the manifold/Langevin flow — which would upgrade the sharpness clause (P.iii) from conditional to unconditional and leave the whole synthesis at theorem grade.

REFERENCES

- [1] I. Drozdov, “Boundaries of stationary feature learning: A minimax barrier for scaling laws and the necessity of compositional structure,” companion theory manuscript (TR), 2025.
- [2] I. Drozdov, “Foundations of a theory of composable abstractions,” companion foundations manuscript (AB), 2025.
- [3] I. Drozdov, “Spectral scaling benchmark: experimental verification of the stationary feature-learning barrier theory,” companion empirical manuscript (BM), 2025.
- [4] R. Schaeffer, B. Miranda, and S. Koyejo, “Are emergent abilities of large language models a mirage?” in *NeurIPS*, 2023.
- [5] J. Wei *et al.*, “Emergent abilities of large language models,” *Transactions on Machine Learning Research*, 2022.
- [6] E. Abbe *et al.*, “SGD learning on neural networks: leap complexity and saddle-to-saddle dynamics,” in *COLT*, 2023.
- [7] G. Yang and E. J. Hu, “Tensor programs IV: feature learning in infinite-width neural networks,” in *ICML*, 2021.

- [8] A. Caponnetto and E. De Vito, “Optimal rates for the regularized least-squares algorithm,” *Found. Comput. Math.*, vol. 7, pp. 331–368, 2007.
- [9] A. B. Tsybakov, *Introduction to Nonparametric Estimation*. Springer, 2009.
- [10] R. A. DeVore, “Nonlinear approximation,” *Acta Numer.*, vol. 7, pp. 51–150, 1998.
- [11] T. Poggio *et al.*, “Why and when can deep—but not shallow—networks avoid the curse of dimensionality,” *Int. J. Autom. Comput.*, vol. 14, pp. 503–519, 2017.
- [12] A. Damian *et al.*, “Neural networks can learn representations with gradient descent,” in *COLT*, 2022.
- [13] C. Davis and W. M. Kahan, “The rotation of eigenvectors by a perturbation III,” *SIAM J. Numer. Anal.*, vol. 7, pp. 1–46, 1970.
- [14] T. Kato, *Perturbation Theory for Linear Operators*. Springer, 1995.
- [15] M. I. Freidlin and A. D. Wentzell, *Random Perturbations of Dynamical Systems*, 3rd ed. Springer, 2012.
- [16] B. Bordelon, A. Atanasov, and C. Pehlevan, “A dynamical model of neural scaling laws,” in *ICML*, 2024.
- [17] J. Hoffmann *et al.*, “Training compute-optimal large language models,” *arXiv:2203.15556*, 2022.